

2-13 Compare 4-Digit Numbers

Learning goal: I can compare 4-digit numbers using a number line or place value, and write $<$, $>$, $=$, or $\neq$.

Textbook: pages 83–86. The book compares 1,324 and 1,249 with a number line and with place value.

Words to Know

compare	Decide which number is greater, which is less, or if they are equal.
greater than ($>$)	$5,362 > 5,318$ means 5,362 is greater than 5,318.
less than ($<$)	$2,371 < 2,436$ means 2,371 is less than 2,436.
equal to ($=$)	The numbers are the same: every place has the same digit.
not equal to ($\neq$)	The numbers are different: at least one place has a different digit.
place value	The value of a digit because of its place: thousands, hundreds, tens, ones.

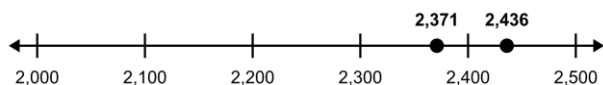
How to Teach It

- Number line:** numbers get greater to the right. Plot both numbers. The one farther right is greater.
- Place value:** line up the numbers. Compare the thousands first. If they are the same, compare the hundreds, then the tens, then the ones.
- Stop at the **first** place where the digits are different. That place decides.
- A 3-digit number has 0 thousands, so any 4-digit number is greater: $3,158 > 958$.
- Read the sentence aloud: “2,814 is less than 2,841.” The open side of $<$ or $>$ faces the greater number.

Tip: Watch for numbers with the same digits in a different order (2,814 and 2,841). They are not equal. Compare place by place.

Worked Examples

Example 1: Use a number line



2,436 is farther right, so $2,371 < 2,436$.

Example 2: Use place value

	Th	H	T	O
5,362	5	3	6	2
5,318	5	3	1	8

Thousands and hundreds are the same. Tens: $6 > 1$, so $5,362 > 5,318$.

Example 3: Equal or not equal

6,207 and 6,270: the tens are different (0 and 7).

So $6,207 \neq 6,270$. Also $6,207 < 6,270$.

Example 4: Different number of digits; order three

3,158 vs 958: think of 958 as 0,958. 3 thousands $>$ 0 thousands, so $3,158 > 958$.

Order 4,572, 3,986, 4,519: $3,986 < 4,519 < 4,572$.

Skills Checklist *Check off each skill your child can do without help.*

- | | |
|---|--|
| ☐ Plot two numbers on a number line and tell which is greater. | ☐ Write $<$ and $>$ correctly and read them aloud. |
| ☐ Compare by place value, starting with the thousands. | ☐ Compare a 3-digit number with a 4-digit number. |
| ☐ Use $=$ and $\neq$ correctly. | ☐ Order three numbers and solve “which is more” problems. |

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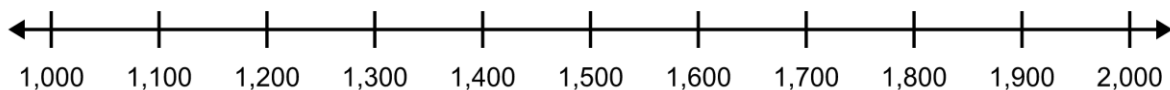
Name _____ Date _____

Together *Work with your grown-up. Talk about each step.*

- A.** Which is greater? Write $<$, $>$, or $=$. Explain your thinking.

6,384 ○ **6,381**

- B.** Plot 1,450 and 1,720 on the number line. Then write $<$, $>$, or $=$.

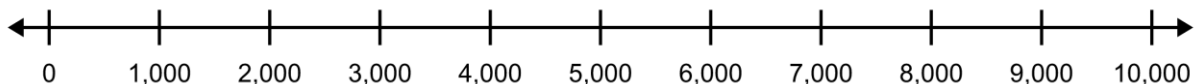
**1,450** ○ **1,720**

- C.** Are the numbers equal or not equal? Write $=$ or $\neq$. Explain.

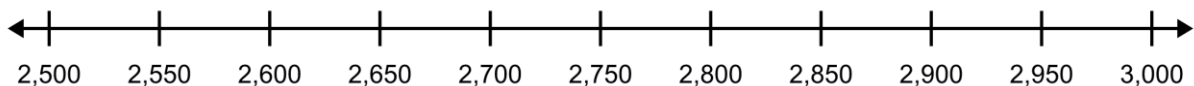
5,269 ○ **5,296**

On My Own *Try these by yourself. Show your work.*

- 1.** Plot each number on the number line. Then write $<$, $>$, or $=$.

6,500 ○ **8,000**

- 2.** Plot each number on the number line. Then write $<$, $>$, or $=$.

2,945 ○ **2,610**

3. Write = or $\neq$. Explain.

7,316 ☐ 7,316

4. Write = or $\neq$. Explain.

4,052 ☐ 4,025

5. Write < or >. Explain.

6,215 ☐ 5,980

6. Write < or >. Explain.

9,412 ☐ 9,705

7. Write <, >, or =.

2,567 ☐ 567

8. Write <, >, or =.

4,728 ☐ 4,782

9. Write <, >, or =.

3,602 ☐ 3,606

10. Write <, >, or =.

836 ☐ 1,059

11. Oak School has 1,850 books. Pine School has 1,792 books. Which school has more books? How do you know?

12. Order these numbers from **least** to **greatest**: 5,302 4,999 5,230

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Together

A	6,384 > 6,381 <i>Thousands, hundreds, and tens are the same. Ones: 4 > 1.</i>
B	1,450 is halfway between 1,400 and 1,500; 1,720 is between 1,700 and 1,800. 1,450 < 1,720 <i>Accept dots placed close to the right spots; 1,720 must be to the right of 1,450.</i>
C	5,269 ≠ 5,296 <i>Thousands and hundreds match, but the tens are different (6 and 9). Also 5,269 < 5,296.</i>

On My Own

1	6,500 halfway between 6,000 and 7,000; 8,000 on its tick. 6,500 < 8,000
2	2,945 just left of 2,950; 2,610 just right of 2,600. 2,945 > 2,610 <i>This line counts by 50s.</i>
3	7,316 = 7,316 <i>Every place has the same digit.</i>
4	4,052 ≠ 4,025 <i>Same digits, different places: the tens are 5 and 2. Also 4,052 > 4,025.</i>
5	6,215 > 5,980 <i>Thousands: 6 > 5. The thousands decide, even though 980 > 215.</i>
6	9,412 < 9,705 <i>Thousands are the same. Hundreds: 4 < 7.</i>
7	2,567 > 567 <i>567 has 0 thousands.</i>
8	4,728 < 4,782 <i>Tens: 2 < 8.</i>
9	3,602 < 3,606 <i>Ones: 2 < 6.</i>
10	836 < 1,059 <i>836 has 0 thousands.</i>
11	Oak School. Thousands are the same (1). Hundreds: 8 > 7, so 1,850 > 1,792.
12	4,999 < 5,230 < 5,302 <i>4,999 has only 4 thousands. For 5,230 and 5,302, hundreds: 2 < 3.</i>